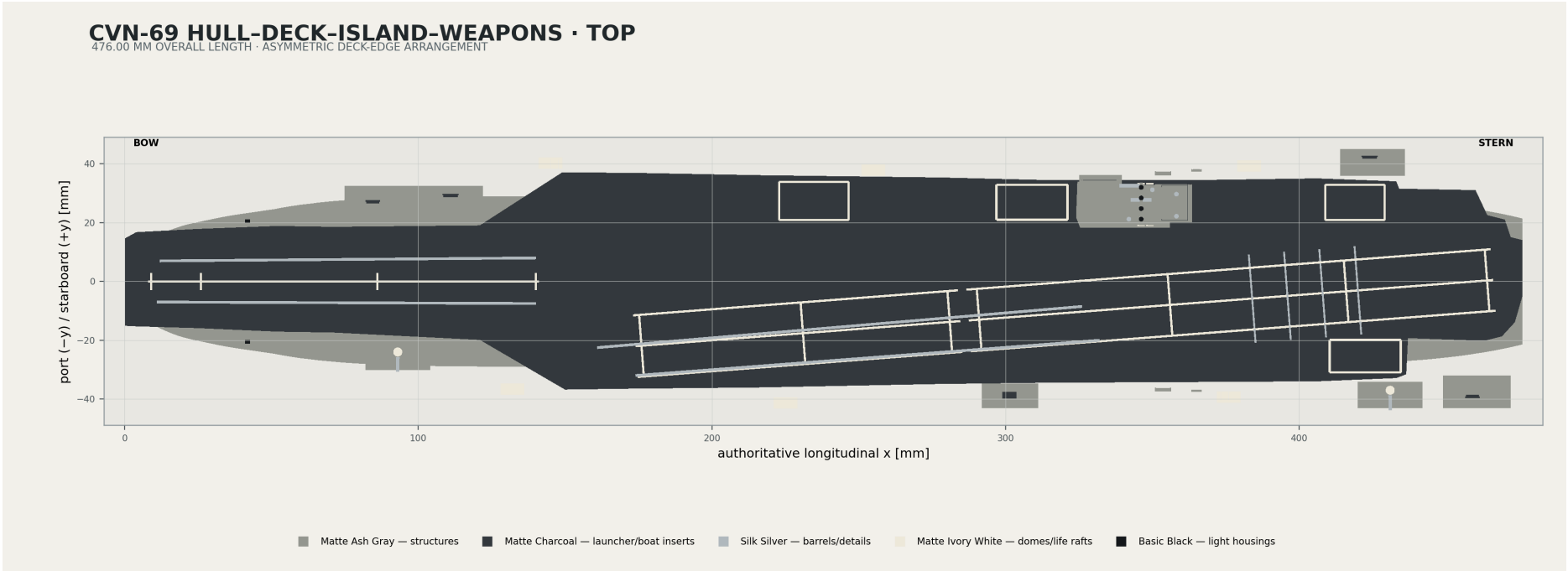


# CVN-69 Defensive Systems and Deck-Edge Drawings

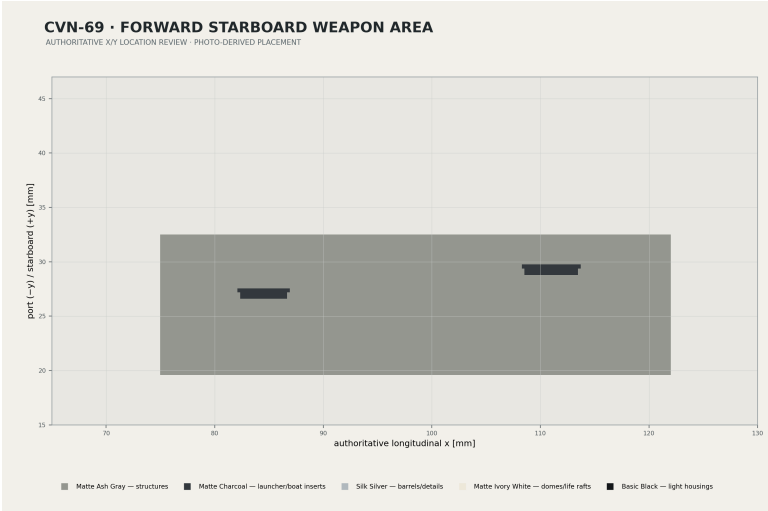
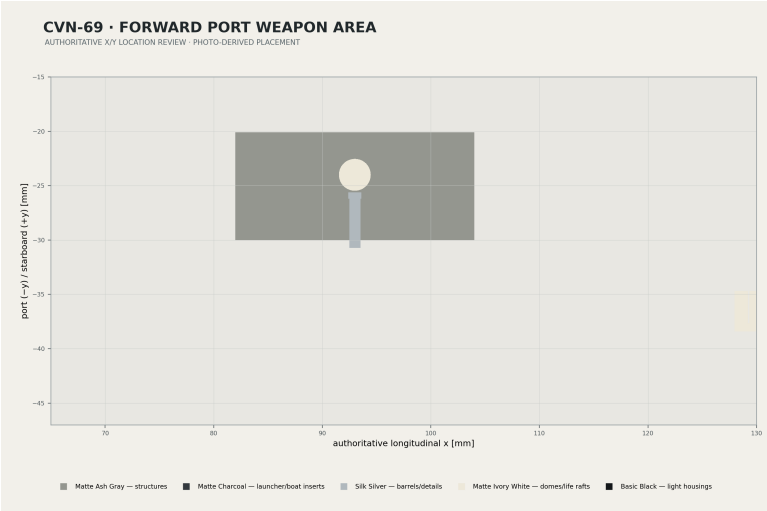
MILESTONE 4 · 1:700 · 2023–2024 DEPLOYMENT FIT · GLUE-ONLY · PUBLIC-PHOTO-DERIVED REVIEW



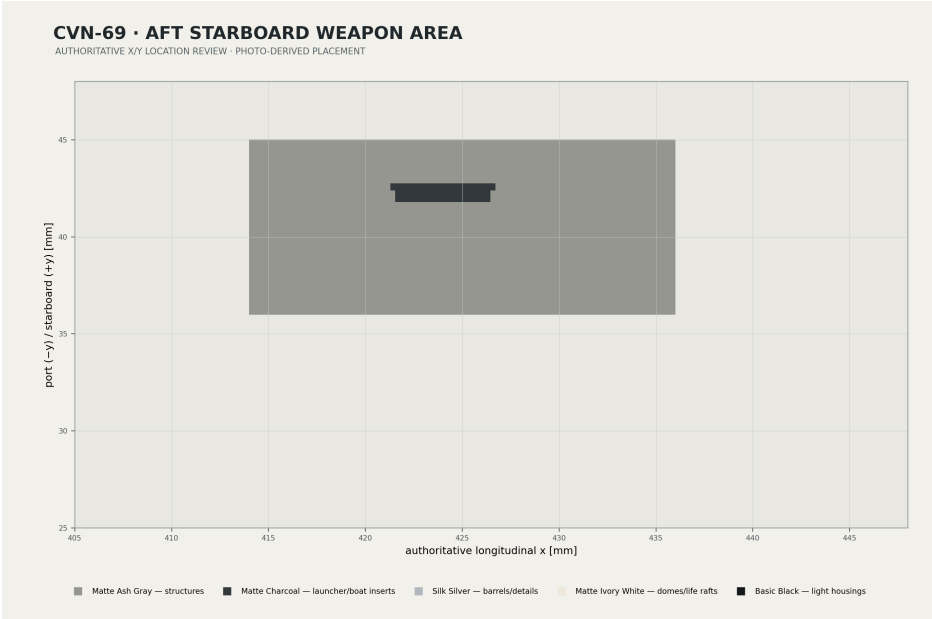
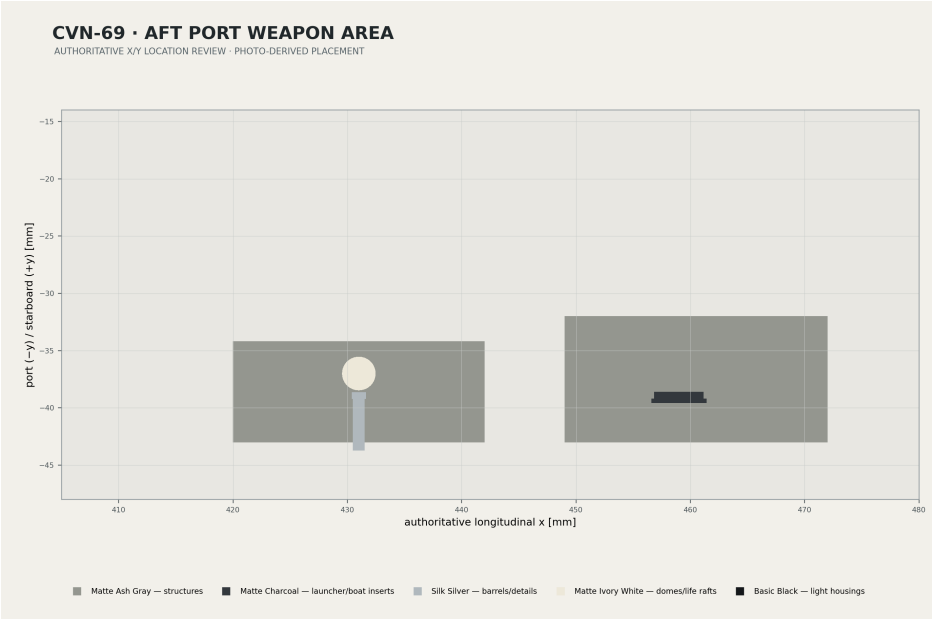
| Datum / count          | Value                                                            | Classification                               |
|------------------------|------------------------------------------------------------------|----------------------------------------------|
| Ship coordinate system | X 0.00 bow → 476.00 stern; port Y negative; starboard Y positive | approved Milestones 1–3                      |
| Deck underside / top   | z = 31.50 / 34.50 mm                                             | approved integration datum                   |
| Defensive fit          | 2 CIWS · 2 RAM · 2 Mk 29 / ESSM                                  | official-system evidence + cross-image count |
| Production objects     | 43 plus 2 coupon objects                                         | new parametric BReps                         |

# Authoritative installation coordinates and footprints

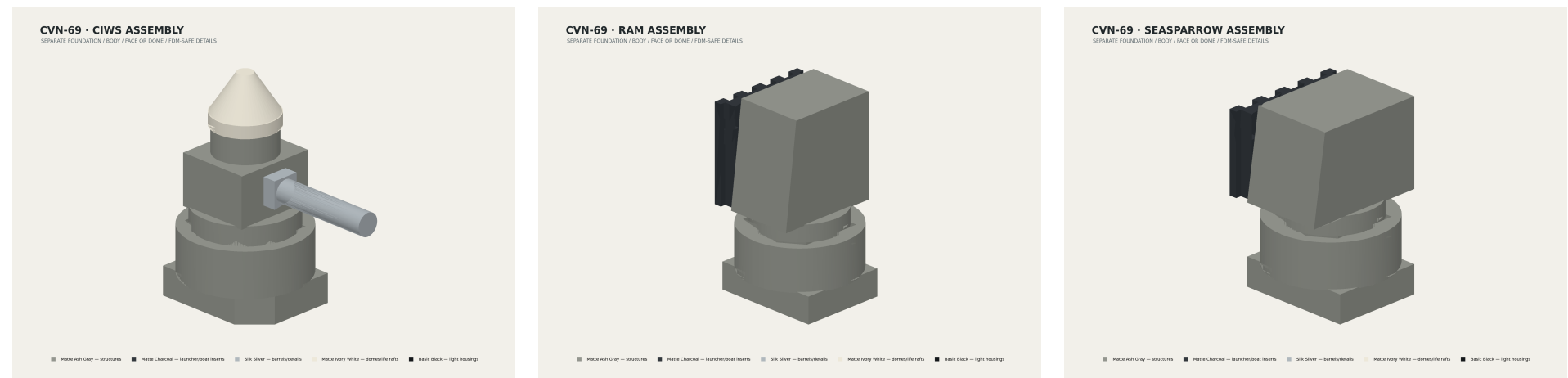
| Installation                    | Family     | X      | Y      | Z seat | Platform                  | Confidence                     |
|---------------------------------|------------|--------|--------|--------|---------------------------|--------------------------------|
| CIWS_01_Forward_Port            | CIWS       | 93.00  | -24.00 | 34.50  | Sponson_Forward_Port      | high family / medium placement |
| RAM_01_Forward_Starboard        | RAM        | 84.50  | 25.00  | 34.50  | Sponson_Forward_Starboard | medium-high                    |
| SeaSparrow_01_Forward_Starboard | SeaSparrow | 111.00 | 27.00  | 34.50  | Sponson_Forward_Starboard | medium-high                    |
| CIWS_02_Aft_Port                | CIWS       | 431.00 | -37.00 | 34.50  | Sponson_Aft_Port_CIWS     | high family / medium placement |
| RAM_02_Aft_Port                 | RAM        | 459.00 | -37.00 | 34.50  | Sponson_Aft_Port_RAM      | medium-high                    |
| SeaSparrow_02_Aft_Starboard     | SeaSparrow | 424.00 | 40.00  | 34.50  | Sponson_Aft_Starboard     | medium                         |



# Aft weapon areas



# Major system dimensions and part breakdown



| Representation  | Major CAD envelope / minimum                               | Print orientation                                           |
|-----------------|------------------------------------------------------------|-------------------------------------------------------------|
| CIWS            | foundation + upper body + dome + fixed 1.00 mm barrel      | foundation key-down; body/dome base-down; barrel z=0 export |
| Mk 49 RAM       | foundation + 5.80 mm launcher + separate 4×3 FDM-safe face | bases down; dark face flat                                  |
| Mk 29 / ESSM    | foundation + 4.80 mm launcher + separate 4×2 FDM-safe face | bases down; dark face flat                                  |
| Common platform | 2.40 mm thick; 1.20 mm socket; 1.20 mm remaining skin      | broad face down                                             |

# Exploded assembly

## CVN-69 MILESTONE 4 · EXPLODED ASSEMBLY

43 PRODUCTION OBJECTS · FOUNDATIONS · WEAPONS · LIFE-SAFETY · DECK-EDGE DETAILS

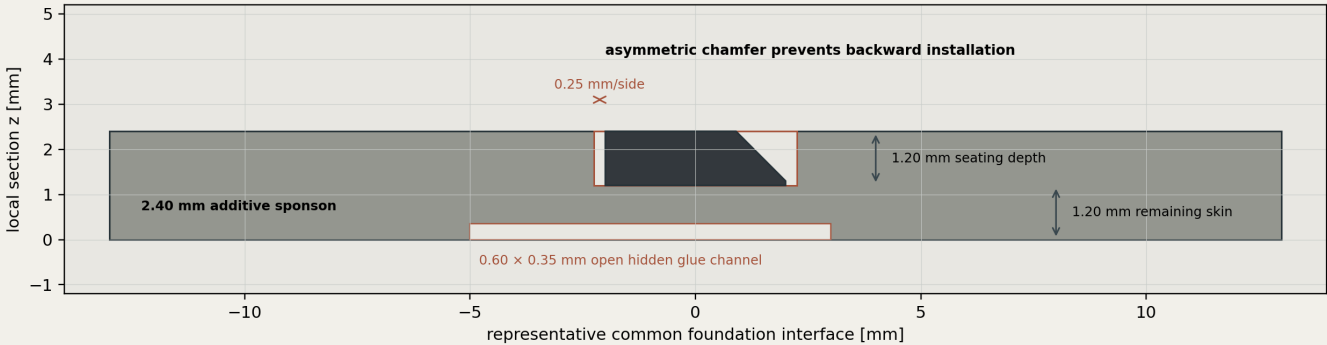


■ Matte Ash Gray — structures   ■ Matte Charcoal — launcher/boat inserts   ■ Silk Silver — barrels/details   ■ Matte Ivory White — domes/life rafts   ■ Basic Black — light housings

# Common glue-only interface section

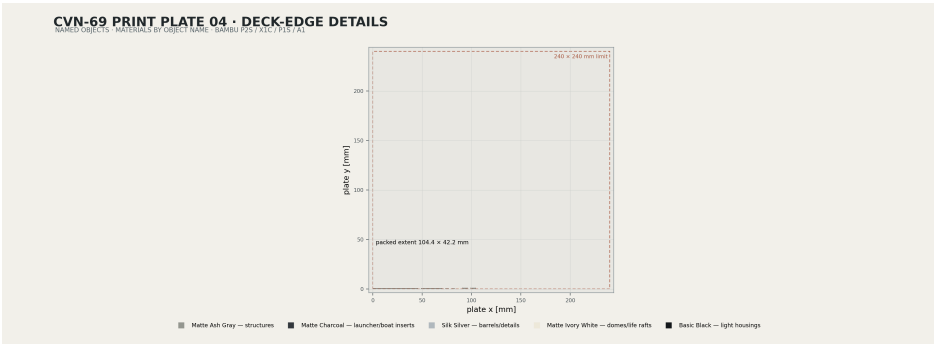
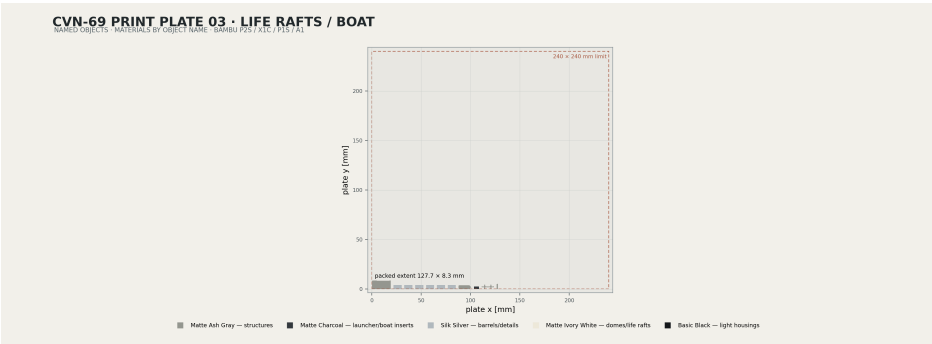
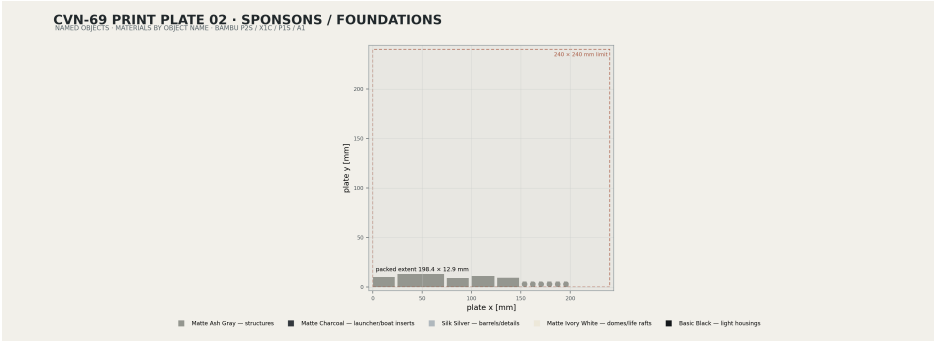
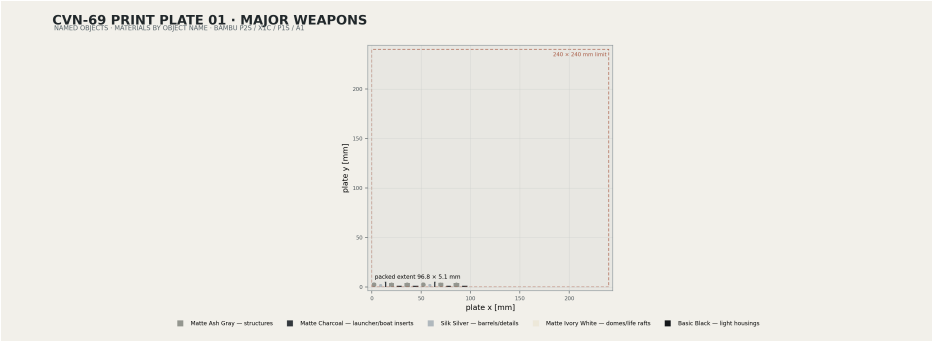
## CVN-69 WEAPON FOUNDATION INTERFACE FAMILY

GLUE-ONLY · 0.25 mm/SIDE · OPEN SOCKET · NO METAL OR PURCHASED CONNECTORS



| Interface parameter     | Production value                | Acceptance                       |
|-------------------------|---------------------------------|----------------------------------|
| Nominal clearance       | 0.25 mm per side                | coupon dry-fit; no forcing       |
| Key / socket depth      | 1.20 mm                         | vertical seating error < 0.10 mm |
| Remaining platform skin | 1.20 mm                         | ≥ 1.20 mm                        |
| Glue channel            | 0.60 × 0.35 mm; open and hidden | no trapped cavity                |
| Installed lean          | 0.00° CAD intent                | < 0.30°                          |

# Print-plate layouts and material assignments



Ash Gray: structures. Charcoal: launcher/boat inserts. Ivory White: CIWS domes and life rafts. Silk Silver: CIWS barrels/details. Basic Black: navigation-light housings. No AMS slot is fixed.